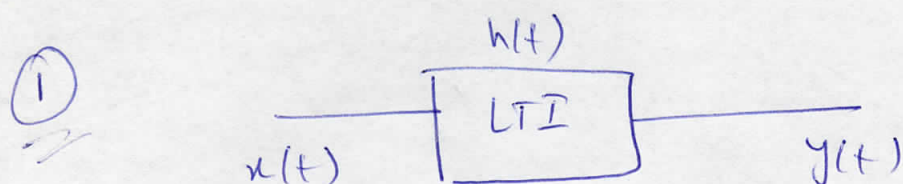


Wrap up introduction to Fourier with the final three points.

- ① What happens to frequencies as they pass through LTI systems?
- ② Fourier series: spectra of periodic signals.
- ③ Link b/w Z , $\mathcal{L}$, and $\mathcal{F}$.



$\Rightarrow$ wkt: $y(t) = x(t) * h(t)$

Take FT $Y(\omega) = X(\omega) H(\omega)$ (recall $H(\omega) = \text{complex} = |H(\omega)| e^{j\angle H(\omega)}$)

$\rightarrow$ so $H(\omega)$ acts as a scaling and phase shifting in the frequencies of $x(t)$ (~~characterized~~ represented by $X(\omega)$)

$\rightarrow$ To see this, write $H(\omega) = |H(\omega)| e^{j\angle H(\omega)}$

and $X(\omega) = |X(\omega)| e^{j\angle X(\omega)}$

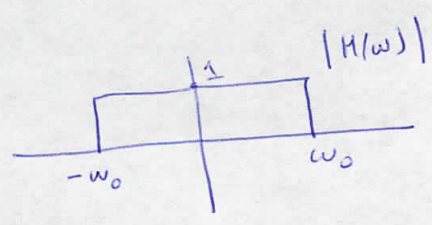
$\rightarrow$ Then output is $Y(\omega) = \underbrace{|H(\omega)| |X(\omega)|}_{\text{amplitude scale}} e^{j(\angle X(\omega) + \angle H(\omega))}$

$|Y(\omega)|$ $\angle Y(\omega)$ phase shifter

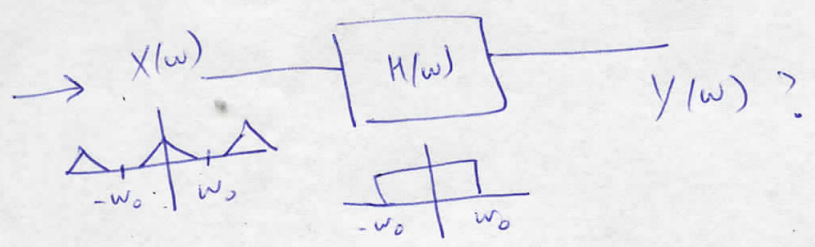
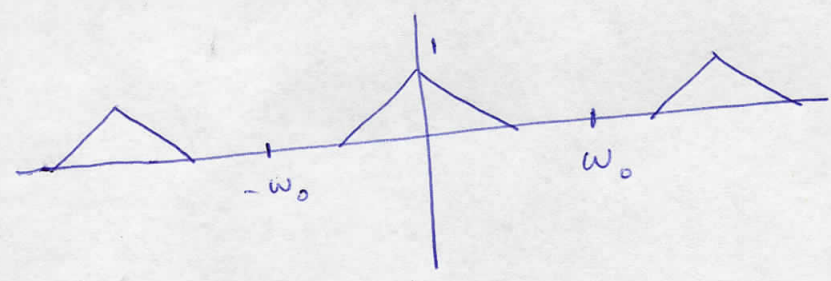
$\rightarrow H(\omega) =$ Transfer function, also called Frequency Response of the system as it tells what the system will do to frequencies.

eg → Let's say there is a system that behaves as a filter, saying I will only allow frequencies below ω_0 to pass and will block all others ('Low Pass Filter')

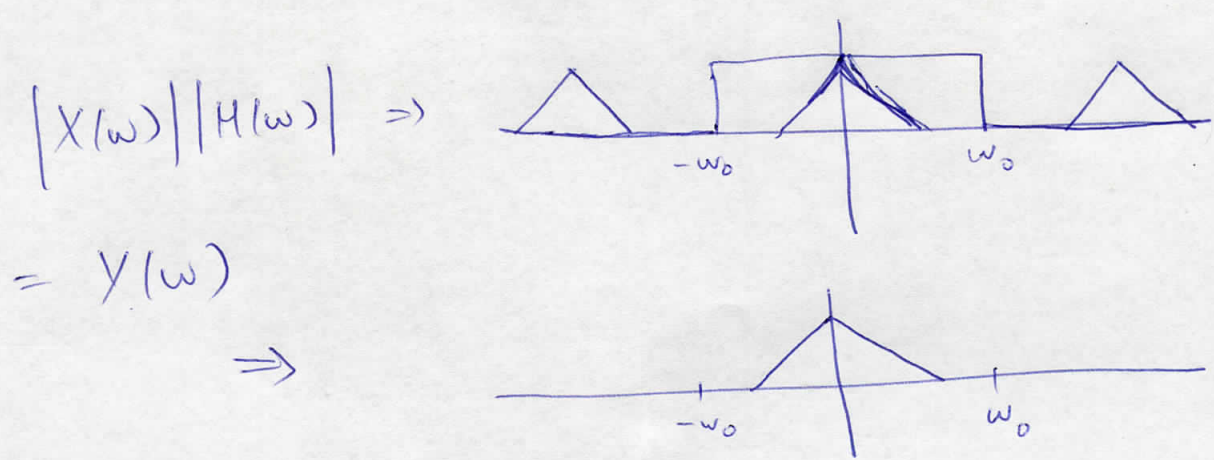
→ The $|H(\omega)|$ of such a system could look like



→ now what happens to a signal that has frequencies both above and below ω_0 ? , e.g., let's say the signal frequencies are



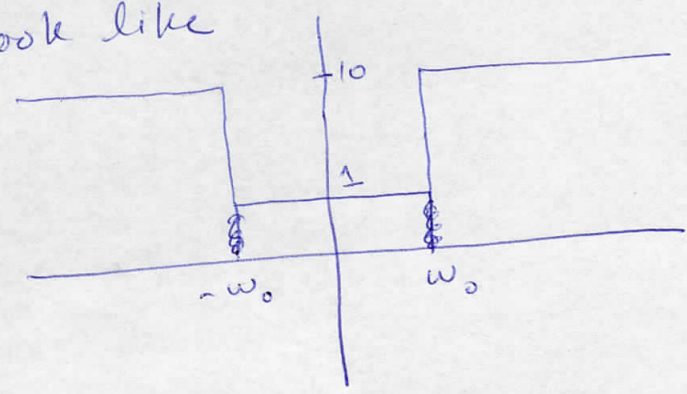
→ since $|Y(\omega)| = |X(\omega)| |H(\omega)|$, we get



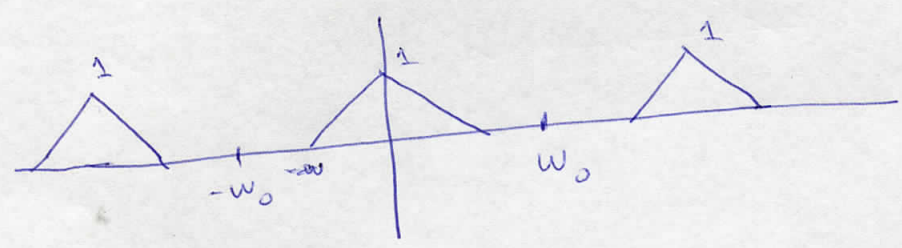
→ i.e only low frequency part remains after filtering.

ex 2 $\Rightarrow$ How about a system that says $\hat{y}$ will let frequencies below ω_0 pass through unchanged but will amplify frequencies above ω_0 by 10x?

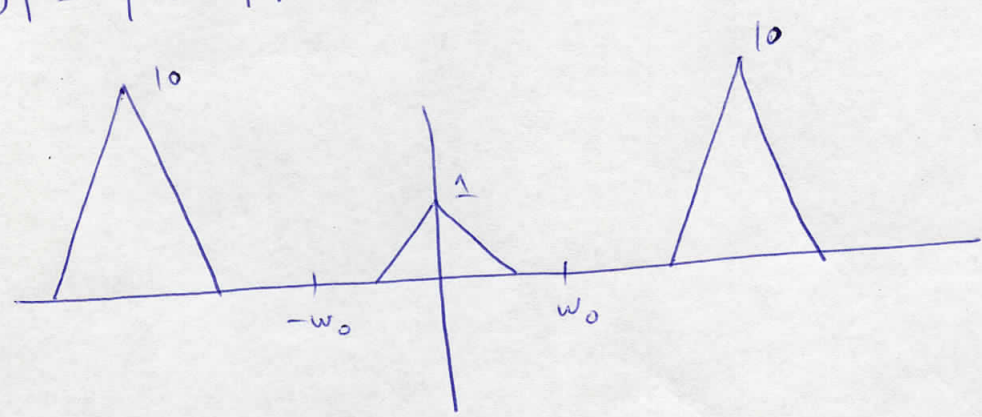
$\rightarrow$ The ^{magnitude} freq. resp. $|H(\omega)|$ of such system (selective amplifier?) may look like



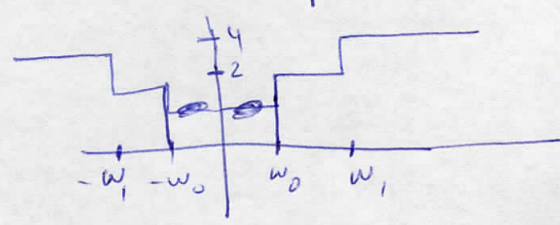
$\rightarrow$ now what happens to freqs of $x(\omega)$ with $|x(\omega)|$



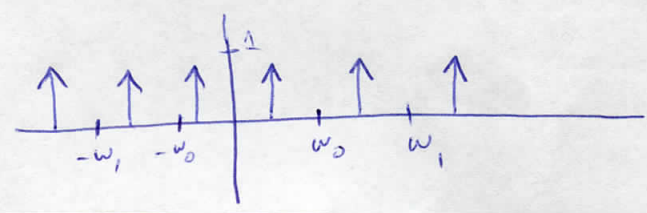
we get $\rightarrow |y(\omega)| = |x(\omega)| |H(\omega)|$



ex 3 Given $|H(\omega)|$

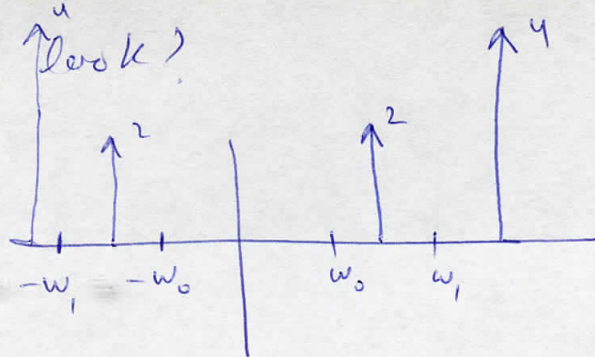


and $|x(\omega)|$

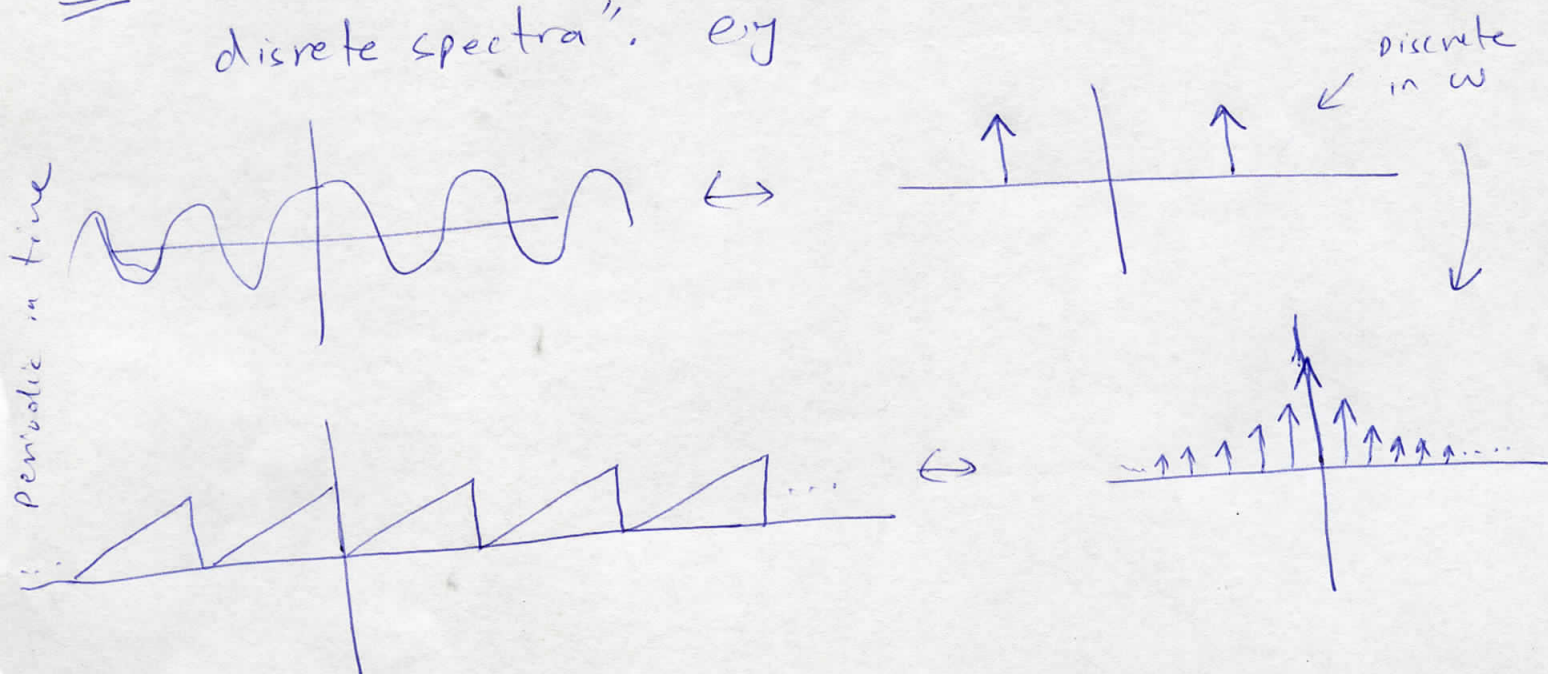


How should $|Y(\omega)|$ look?

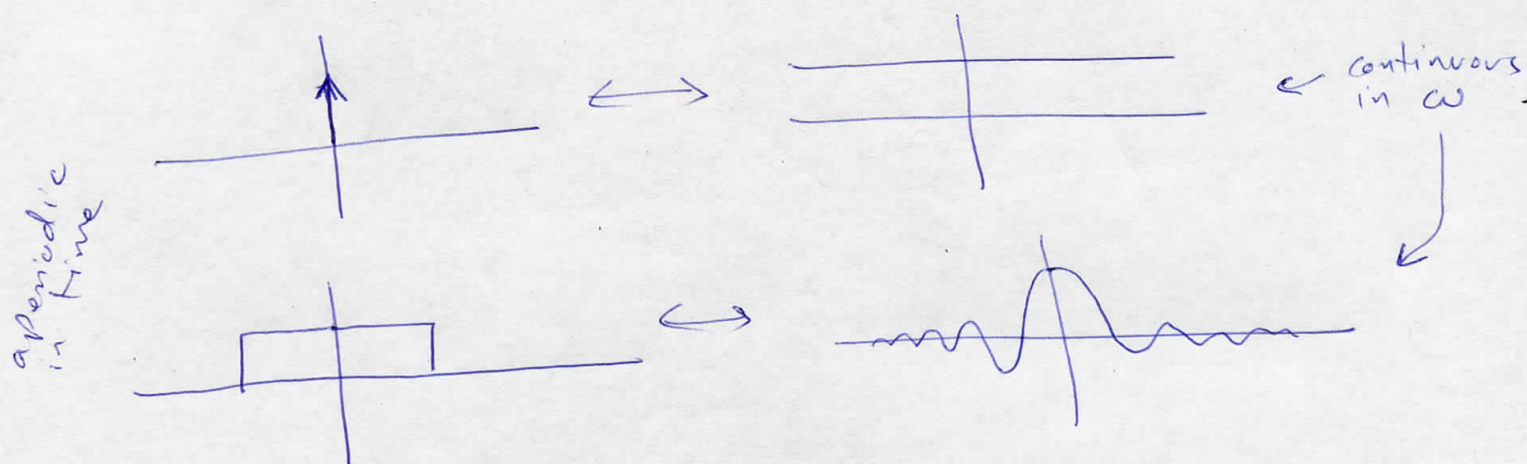
$\rightarrow |Y(\omega)| \Rightarrow$



② Previously we saw that: "Periodic signals have discrete spectra". eg



whereas aperiodic signals have continuous spectra, eg



$\rightarrow$ Discrete functions lead to series, so instead of having

$$x(t) = \frac{1}{2\pi} \int X(\omega) e^{j\omega t} d\omega \leftarrow \text{Fourier Integral (Transform)}$$

we can have something like

$$x(t) = \sum_{\forall n} X_n e^{jn\omega_0 t}$$

← Fourier Series
in all practical cases

→ In fact, it can be shown that a periodic signal with ^{fundamental} period T_0 and angular frequency $\omega_0 = \frac{2\pi}{T_0}$ can be written as a sum of sinusoids of frequencies ω_0 and its multiples ($2\omega_0, 3\omega_0, 4\omega_0, \dots$), i.e. it can be written in terms of

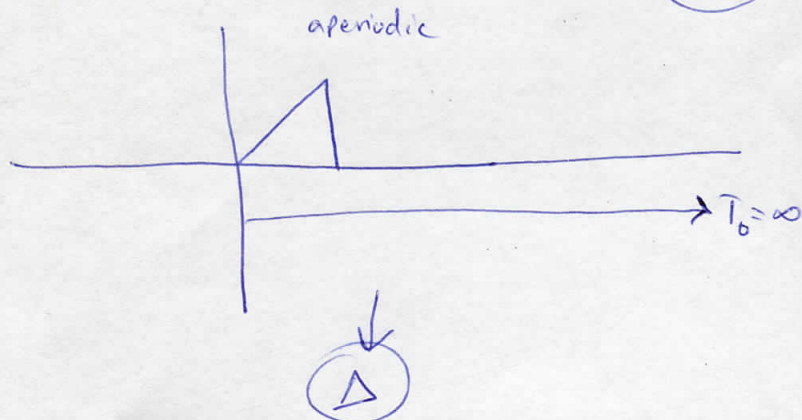
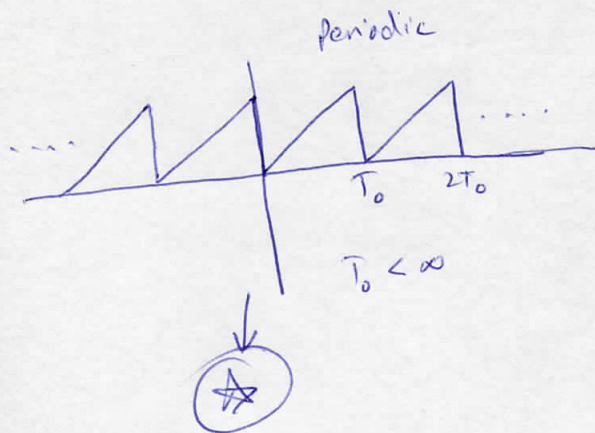
$$e^{j\omega_0 t}, e^{j2\omega_0 t}, e^{j3\omega_0 t}, \dots, e^{jn\omega_0 t}, \dots$$

→ This is called the Exponential Fourier series, and it says that $x(t)$ is periodic with fundamental frequency ω_0 then we can write it as

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \quad \text{--- } \textcircled{\star}$$

→ In fact it can be shown that as ~~the~~ $T_0 \rightarrow \infty$ (i.e. signal becomes aperiodic) $\textcircled{\star}$ becomes the Fourier Transform (integral)

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad \text{--- } \textcircled{\Delta}$$



⑥

→ Furthermore, we can also show that $x(t)$ with fund. freq. ω_0 can be written in terms of $\cos(n\omega_0 t)$ and $\sin(n\omega_0 t)$ as

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

→ This representation is called
"Trigonometric Fourier Series".

→ we will not touch upon these in further detail.

③ $\mathbb{Z}, \mathcal{L}, \mathcal{F}$

we have seen that $\mathbb{Z}, \mathcal{L}$, and $\mathcal{F}$ use bases functions

$\mathbb{Z}^n, e^{st}$, and $e^{j\omega t}$ respectively with (z & s complex numbers)

such that

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad s = \sigma + j\omega$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

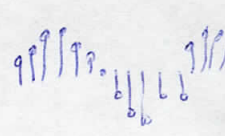
→ How are they related?

(7)

$$z = e^{sT}$$

Discrete signal
(case with
signal sampled at
period T)

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) e^{-snT}$$

ingredients: 

Laplace

- continuous signals
- uses complex exponentials allowing for pure, growing, and decaying sinusoids as

$$e^{-st} = e^{-\sigma t} e^{-j\omega t}$$

Ingredients
limited to
pure complex
sinusoids only
 $e^{-j\omega t}$ ($\sigma=0$)

(*)

Fourier
Tx.

Ingredients:



Ingredients:



$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

(*) Note that even though we may get FT from LT by setting $s = j\omega$, the two Tx's may have different ROCs.